Kanhegaonkar Malhar Ganesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.07
Cambridge Vidyalaya, Nanded	Class XII (Maharashtra State Board)	2024	82.00%
Cambridge Vidyalaya, Nanded	Class X (Maharashtra State Board)	2022	97.60%

SCHOLASTIC ACHIEVEMENTS

- **Received the Academic Excellence Award (2025)** at IIT Hyderabad for securing the **highest SGPA** in Semester I among the Engineering Science batch.
- **Ranked among the top 4 students** in the Engineering Science cohort, securing selective admission into the **Computer Science specialization**.

SKILLS

- **Programming Languages** - C, C++, Python
- **Fundamentals** - Data Structures and Algorithms, Problem Solving, Object-Oriented Programming (OOP), STL
- **Tools & Technologies** - Git, GitHub, Linux, Solid Edge, Latex, NumPy

RELEVANT COURSES

- **Computer Science** - Introduction to Programming, Discrete Mathematics for Computer Science, Data Structures and Applications, Algorithms, Theory of Computation, Operating Systems 1, Artificial Intelligence
- **Mathematics** - Probability and Random Variables, Calculus 1, Calculus 2, Introduction to Statistics, Complex Variables, Differential Equations, Transform Techniques

PROJECTS

TCP Echo Client/Server with IPv4/IPv6 Multicast | *C, TCP/IP, Sockets*

- Developed a C-based **TCP echo client/server** using POSIX socket APIs, supporting **hostname resolution via getaddrinfo()** and dual-stack IPv4/IPv6 communication.
- Implemented **IPv4 and IPv6 multicast** to distribute server-received messages to multiple clients, with I/O multiplexing using select().
- Added **TCP RTT measurement and timestamped network event logging** to evaluate communication latency and simplify debugging

Adaptive Mental Math Trainer | *C++20, CMake, OOP, STL*

- Developed a terminal-based **Adaptive Mental Math Trainer** using **C++20**, **Object-Oriented Programming (OOP)**, STL, and **nlohmann/json** to provide personalized arithmetic practice with persistent progress tracking.
- Implemented an adaptive question selection algorithm that prioritizes weaker skills using **accuracy** and **response-time** metrics, while supporting prerequisite-based skill progression.
- Designed a modular architecture using **polymorphism** and the **Single Responsibility Principle (SRP)**, enabling reusable question generators and straightforward addition of new arithmetic skills.
- Built per-skill analytics with accuracy, response time and streak tracking, persisting user performance across sessions.

Producer-Consumer Synchronization | *C, System V IPC, Semaphores, Linux*

- Implemented a **Producer-Consumer** system in C using **System V Shared Memory** and **Semaphores**, enabling synchronized communication between multiple processes through a bounded **Circular Buffer**.
- Designed **atomic multi-semaphore synchronization** using **semop()** with **Mutex**, **Empty**, and **Full** semaphores to

ensure **mutual exclusion** and prevent **buffer overflow/underflow**.
- Managed **Inter-Process Communication (IPC)** by integrating **fork()**, **shmget()**, **shmat()**, and proper **IPC resource cleanup**, ensuring safe concurrent execution across producer and consumer processes.

Probability Simulations | *Python, NumPy, Matplotlib*

- Implemented **Monte Carlo** algorithms including **inverse CDF transform**, **rejection sampling**, **importance sampling**, and the **Box-Muller transform** for random variable generation from first principles using only uniform random variables.
- Developed statistical simulations to verify the **Central Limit Theorem**, generate empirical CDFs, and validate theoretical probability distributions through numerical experiments.
- Applied **importance sampling** for rare-event probability estimation, demonstrating significantly lower estimator variance and improved reliability compared to direct Monte Carlo simulation.

Numerical Nexus | *Python, NumPy, SciPy (Numerical Methods Team Project)*

- Implemented numerical algorithms for **Harshad number** analysis, modified **Legendre polynomials**, **LU decomposition** with partial pivoting, and **Newton-Raphson** root finding to solve computational mathematics problems.
- Contributed to the mathematical theory and algorithm design in a full-stack development of an interactive platform for presenting and visualizing numerical method solutions.

EXTRACURRICULAR

- **Secured 3rd Place** in **AI Day'25 Ideathon**, IIT Hyderabad, for developing an innovative AI-based solution.
- Volunteered with the **National Service Scheme (NSS)** at IIT Hyderabad, contributing to community service and social outreach initiatives.